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(SECP3843) SPECIAL TOPIC IN DATA ENGINEERING

SECTION 02

GROUP 3

TUTORIAL MS AZURE

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1.0 Project Overview

This project aims at establishing an end-to-end data engineering pipeline on the Microsoft Azure services. The overall aim is to model the process of data collection, processing and analysis in real-life organizations to facilitate superior decision-making.

In modern data-driven society, businesses produce a great quantity of data in various forms. Raw data, however, is generally sloppy and not directly useful. This project shows how unstructured data can be processed systematically to extract valuable insights using a managed set of data processing steps.

The solution is based on a modern data architecture model, according to which data are initially captured in a source system and then stored in a data lake and then subsequently processed to various layers (Bronze, Silver and Gold) and finally viewed via a dashboard tool. This stratified methodology assists in enhancing the quality, structure and usability of data.

This project is covered by the scope that includes:

- Information taken in by a source system
- A cloud-based data lake data storage
- Data processing and transformation
- Data analysis and visualization

On the whole, this project represents a practical industry situation, where the companies use cloud technologies to store and process big data effectively.

2.0 Description on the Solution

The proposed solution presents a comprehensive cloud-based data processing system built using Microsoft Azure to efficiently manage the data flow from source to visualization. The objective of this solution is to transform raw and unstructured data into structured, reliable and analysis-ready information through an automated workflow.

The solution is based on the logical sequential processes that begin with the ingestion of data, ensuring that the data is extracted from the source system into the cloud environment. Azure Data Factory is used to orchestrate and automate data movement across different environments. Its integration capabilities make it suitable for connecting the on-premises data sources to Azure services.

After that, those ingested data will be stored in Azure Data Lake Storage Gen2, which is highly scalable, cost-efficient and capable of handling large amounts of structured and unstructured data. To ensure better data management and quality control, a medallion architecture is applied in the transformation process via Azure Databricks. It organizes data into three different layers. In the Bronze layer, raw data is stored in its original state to preserve data integrity. In the Silver layer, the data is cleaned and standardized for consistency. In the Gold layer, aggregated and business-ready data is optimized for analysis. This layered design is adopted to enhance data reliability, consistency and traceability for further processing.

After the transformation stage, the processed data is prepared for analytical querying using Azure Synapse Analytics. It enables efficient data access and supports structure querying, which allows retrieving relevant information quickly and supports data-driven decision-making.

Last but not least, visualization data using Microsoft Power BI, providing an user-friendly and interactive dashboard for understanding the insights. The entire workflow is automated and scheduled to run regularly, ensuring that the data remains up-to-date while minimizing manual operation.

Overall, the solution demonstrated a well-structured and efficient approach by integrating multiple Azure services. The use of the medallion architecture further strengthens the design by improving

data quality and supporting scalable data processing, making the solution suitable for real-world data engineering applications.

3.0 Technology Use

A number of Microsoft Azure services are used in this project, with each tool having a particular role in the data pipeline. The technologies are all integrated to establish an efficient and smooth flow of data.

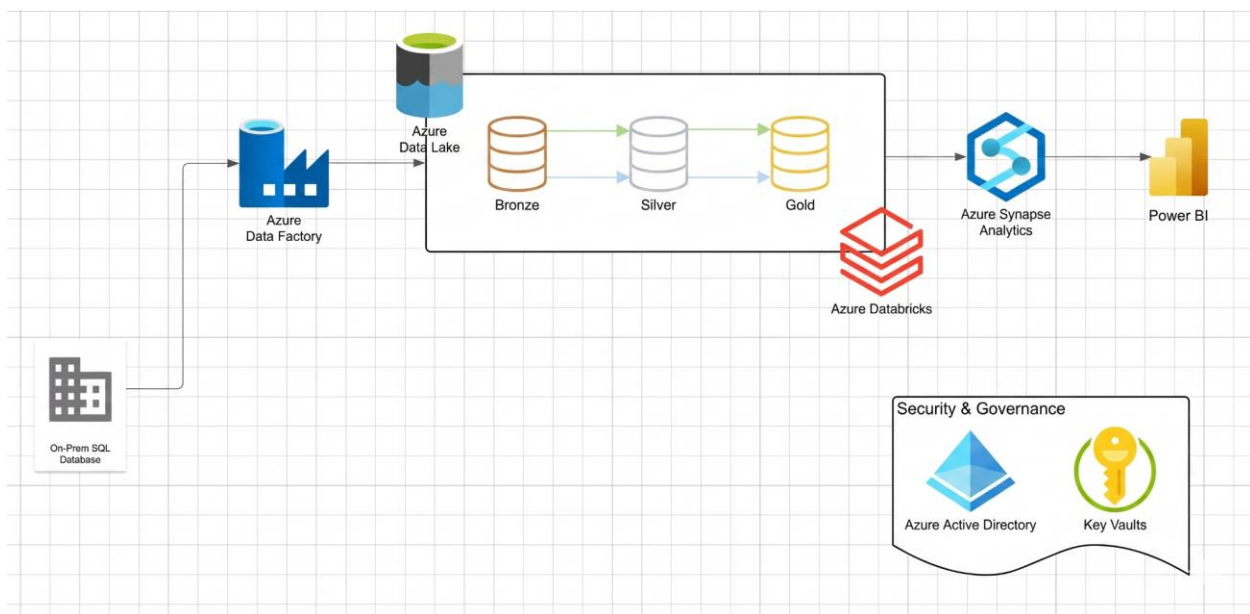


Figure 1: Integrated Microsoft Azure Data Toolchain

3.1 Azure Data Factory

Azure Data Factory is used to handle data ingestion. It performs as a pipeline manager that transfers information of the source system (an on-premises SQL database) into the cloud. It allows scheduling and automation, making the data transfer process reliable and repeatable.

3.2 Azure Data Lake Storage

Azure Data Lake is used as the main storage system for this project. It stored data in various layers:

- Bronze layer: raw data as it was originally
- Silver layer: processed and refined data
- Gold layer: polished data, to be analyzed

This stratified design is used to structure data and makes their processing and analysis easier in the future.

3.3 Azure Databricks

Data processing/data transformation is done with Azure Databricks. It aids in cleaning, filtering and transforming raw data into a more usable form. The tool is robust in that it is capable of processing big data and is scalable in the transformation of data.

3.4 Azure Synapse Analytics

Azure Synapse Analytics is used for querying and analyzing the processed data. It serves as an interface between processed data and visualization layer enabling users to execute analytical queries effectively.

3.5 Power BI

Dashboard and visualizations are created using power BI. It links to the processed data and displays it in an easy to understand manner in the form of charts and reports. This assists users to comprehend clearly trends and insights of the data.

3.6 Key Vault

Key Vault is used to manage security. Sensitive information such as passwords and keys are stored in a secure place in the key vault. This makes the system secure and adhering to good data governance practices.

4.0 Data Pipeline Architecture

The data pipeline architecture is developed based on a layered approach to ensure clarity, efficiency, and scalability in the Azure ecosystem.

To start with the data source layer, the raw data will be collected from the on-premises SQL database. This will serve as the initial input for the pipeline. Now move on to the next layer, which is data ingestion. It uses Azure Data Factory to extract and load data into the cloud environment, which ensures secure and efficient data transfer from on-premises systems to Azure storage.

The next step is the data transformation layer, where raw data will be processed following the medallion architecture by Azure Databricks. This architecture consists of three layers, which are bronze, silver and gold. The Bronze layer stores raw data that is ingested directly from source without modification. Proceeding to the Silver layer, it performs data cleaning and filtering to improve data quality. Finally, the third layer, Gold, holds business-ready data that is suitable for analysis.

Subsequently, the processed data is then passed to the data serving layer using Azure Synapse Analytics. It is efficient for querying and supports large-scale analytics. Finally, the analytics and visualization layer enables users to interact with the data through dashboards. Tools such as Power BI are used to show insights in a clear and user-friendly way, such as sales performance, product trends, and gender-based analysis in this project. This architecture ensures a seamless flow of data from source to insight, while maintaining performance, scalability, and ease of use for users.

5.0 Power BI

5.1 Dashboard Overview

The Power BI dashboard titled “Sales Performance Dashboard”, aims to provide a clear representation of sales data through visualization. It allows analyzing KPIs, assessing the performance of certain products, as well as exploring the distribution of customers among themselves. The dashboard is connected to the Azure SQL Database to ensure structuring and reliability of analyzed data.

The primary purpose of this dashboard is to support data-driven decision making by transforming raw data into meaningful insights through visual representations. By presenting the data in an interactive format, users are able to explore and interpret the information more effectively.

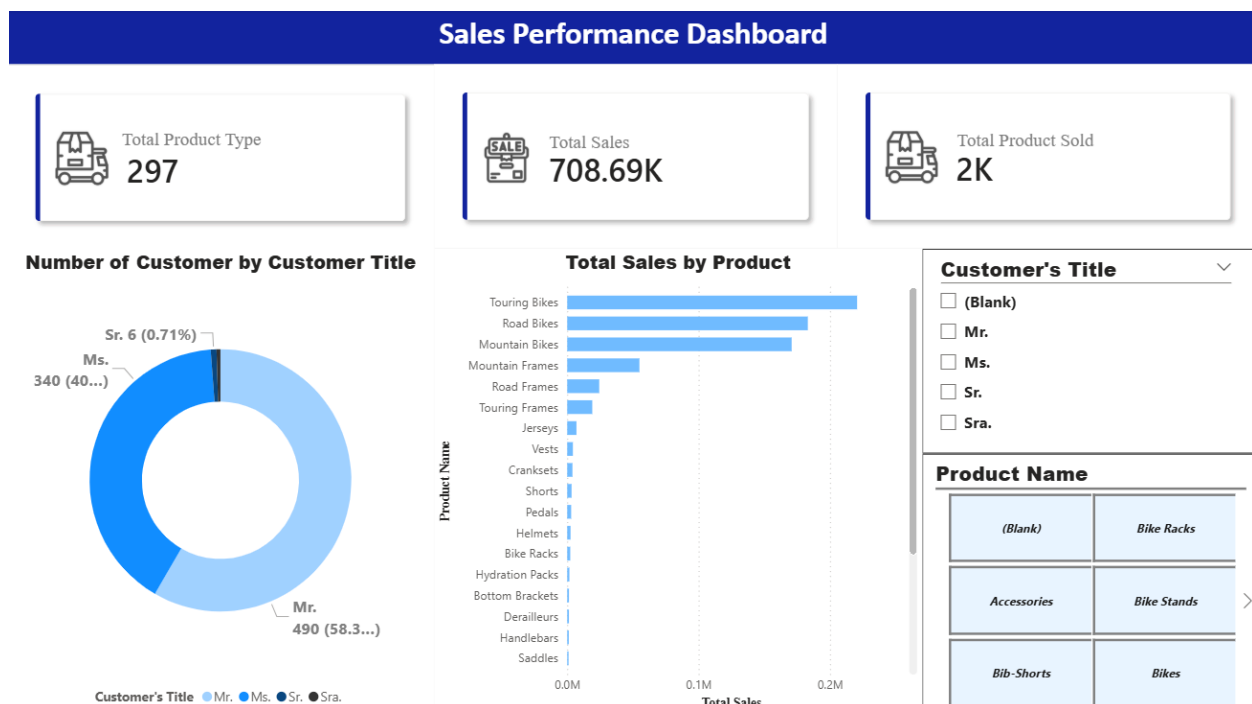


Figure 2: Sales Performance Dashboard

5.2 Key Performance Indicators (KPIs)

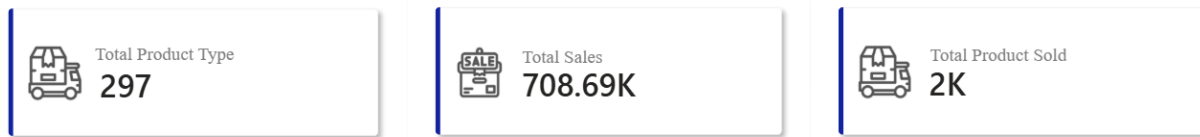


Figure 3: Key Performance Indicators

Inside the dashboard have included three primary KPIs at the top to provide a quick overview of overall performance:

- Total Product Type (297): the total number of product types presented in the dataset
- Total Sales: total revenues from sales as one of the major indicators
- Total Product Sold: the total number of sold products

By having these indicators, it helps users quickly estimate the level of performance before delving deeper into the analysis.

5.3 Data Visualization and Analysis

5.3.1 Number of Customers By Customer Title

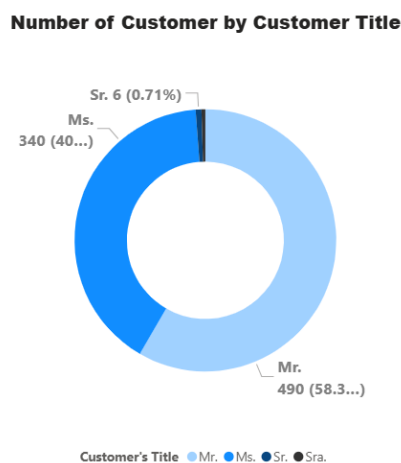


Figure 4: Number of Customer by Customer Title

Figure 4 shows a donut chart that used to present the customer titles such as Mr., Ms., Sr. and Sra. and shows how they are distributed.

Based on the results, it is clear that most of the customers belong to the “Mr.” category; secondly come the customers of “Ms.” type. Other categories only have a minor part of customers’ total number. The obtained results demonstrate that the distribution of customers is quite uneven as it mostly includes representatives of just two categories, Mr. and Ms.

5.3.2 Total Sales by Product

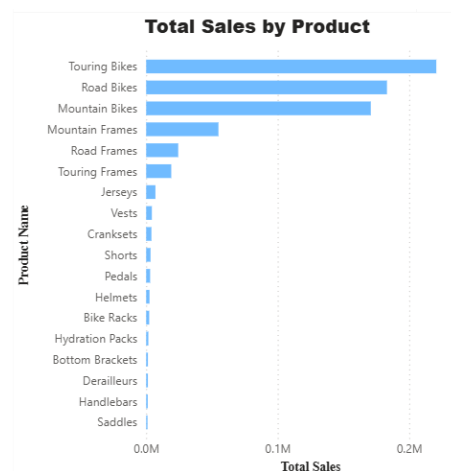


Figure 5: Total Sales by Product

In Figure 5, Bar chart provides information about the total sale per different product types, allowing to easily compare the sales performance over products.

According to the graph, Touring Bikes generate the highest total sales, followed by Road Bikes and Mountain Bikes; accessories and other items have the least sales amount. For insights, one can assume that high-value products have an important impact on total sales, whereas small products are associated with fewer revenues individually.

5.4 Interactive Features – Slicers

The image shows a UI for interactive filters. The first section, 'Customer's Title', has a dropdown arrow and five radio button options: (Blank), Mr., Ms., Sr., and Sra. The second section, 'Product Name', contains a 3x2 grid of buttons. The buttons are labeled: (Blank), Bike Racks, Accessories, Bike Stands, Bib-Shorts, and Bikes. A right-pointing arrow is located to the right of the 'Bike Stands' button.

Customer's Title	
☐ (Blank)	
☐ Mr.	
☐ Ms.	
☐ Sr.	
☐ Sra.	

Product Name	
(Blank)	Bike Racks
Accessories	Bike Stands
Bib-Shorts	Bikes

Figure 6: Interactive Slicer

The dashboard also includes two interactive slicers located on the right-hand side, intended to filter data by Customer's Title and Product Name. These slicers allow users to filter the data dynamically based on selected criteria.

By using the Product Name slicer, users can select products to analyze their sales performance by looking at the KPIs. Moreover, users can also filter customer groups to observe purchasing behavior more closely using the Customer's Title slicer.

This feature helps to enhance the user experience by enabling customized data exploration, making the dashboard more flexible and insightful.

5.5 Data Storytelling and Insights

The dashboard design leads the user through a logical flow of information. It begins with an overview through KPIs, followed by a detailed analysis using charts and concludes with deeper exploration using filters.

Several key insights from the dashboard:

- Sales performance is strongly influenced by high-value products, particularly bicycles
- Customer distribution is concentrated within a few dominant categories

- Product-level analysis highlights both top-performing and lower-performing items

By structuring the information in this way, the dashboard effectively communicates a clear narrative, helping users move from general understanding to more detailed insights.

5.6 Summary

In conclusion, the Power BI dashboard displays the sales data clearly and effectively in a structured way. The addition of well-designed visual elements and insightful analytics, together with interactive features, contributes to the efficiency of the system. Therefore, the effectiveness of Power BI as a business intelligence tool becomes evident.

6.0 Reflection

GUI KAH SIN

This is my first experience using Microsoft Azure, and it was both challenging and insightful. I faced multiple difficulties throughout the whole workflow, such as server connection failures and incorrect pipeline naming. However, through continuous trial and error, I gradually developed an understanding of how a complete data pipeline operates from data extraction and ingestion using Azure Data Factory, to data storage, and finally to data visualization using Microsoft Power BI. One key insight I gained is that each stage of the pipeline is highly interconnected, meaning that even a small misconfiguration in one component can affect the entire workflow. This required careful attention to detail and strengthened my problem-solving skills. Additionally, Power BI appears to be user-friendly, but I realized that creating a visually appealing and effective dashboard requires time, design consideration, and iterative refinement. Overall, this experience has enhanced my understanding of cloud-based data engineering and improved my ability to adapt and learn new technologies independently, which will be valuable for my future career development.

SABRINA HENG WEI QI

This is my first time exposure to Microsoft Azure as I was involved in all three stages of the data engineering process, which involved data storage, building the pipeline and visualization of data. At the beginning stage, I found that it was challenging due to unfamiliarity with the interface and the services. After several practices and learning, I started to understand how the flow of data works from Azure Storage to Azure Data Factory, Azure SQL Database and finally to Power BI. In addition, I learned how to build pipelines and create a dashboard also. To conclude, this project allowed me not only to increase my knowledge about cloud-based data engineering but also improved my skills in solving problems. As a result, I have acquired valuable practical experience that will be very useful for my future education and career.

WOO CHENG SHUAN

The project is the first time that I have worked with Microsoft Azure tools and it has been an interesting and valuable experience. Initially, I struggled to grasp the various services, but through the tutorial, end-to-end, I learned over time how each element of the data pipeline inter-relates within a complete data pipeline. I had practical knowledge on data ingestion, storage, processing and visualization and this allowed me to learn more about the practicalities of data engineering in the real world. I think this information will come in handy when I do my future WBL project with PPG and will also be useful when I train in the industry because cloud technologies such as Azure are common in the industry.

7.0 Reference

Detailed report



01/04/2026 - 30/04/2026

Total: 30:25:08 Billable: 30:25:08 Amount: 0.00 USD

Date	Description	Duration	User
20/04/2026	Write 1.0 & 3.0 Group 3 Section 02 - Reporting	01:41:00 22:07:00 - 23:48:00	Woo Cheng Shuan 0.00 USD
18/04/2026	Learn Azure Group 3 Section 02 - Online learning Azure	04:44:15 21:22:00 - 02:06:15 ⁺¹	Woo Cheng Shuan 0.00 USD
18/04/2026	Write 2.0 & 4.0 Group 3 Section 02 - Reporting	02:06:00 19:47:00 - 21:53:00	GUIKAHSIN 0.00 USD
18/04/2026	Write 5.0 Power BI Group 3 Section 02 - Reporting	01:19:24 09:00:00 - 10:19:24	Sabrina Heng 0.00 USD
17/04/2026	Learn Azure Group 3 Section 02 - Online learning Azure	03:14:00 21:23:00 - 00:37:00 ⁺¹	GUIKAHSIN 0.00 USD
17/04/2026	Learn Azure Group 3 Section 02 - Online learning Azure	01:55:00 13:14:00 - 15:09:00	Woo Cheng Shuan 0.00 USD
17/04/2026	Learn Azure Group 3 Section 02 - Online learning Azure	01:35:00 12:33:00 - 14:08:00	GUIKAHSIN 0.00 USD
17/04/2026	Design Power BI Group 3 Section 02 - Reporting	01:25:49 09:00:00 - 10:25:49	Sabrina Heng 0.00 USD
16/04/2026	Learn Azure Group 3 Section 02 - Online learning Azure	03:23:00 20:33:00 - 23:56:00	GUIKAHSIN 0.00 USD
15/04/2026	Learn Azure Group 3 Section 02 - Online learning Azure	02:24:18 22:18:42 - 00:43:00 ⁺¹	Woo Cheng Shuan 0.00 USD
15/04/2026	Learn Azure Group 3 Section 02 - Online learning Azure	00:16:14 10:20:00 - 10:36:14	Woo Cheng Shuan 0.00 USD
15/04/2026	Learn Azure Group 3 Section 02 - Online learning Azure	01:22:03 09:00:00 - 10:22:03	Sabrina Heng 0.00 USD
14/04/2026	Learn Azure Group 3 Section 02 - Online learning Azure	01:24:11 09:00:00 - 10:24:11	Sabrina Heng 0.00 USD
13/04/2026	Learn Azure Group 3 Section 02 - Online learning Azure	01:18:00 21:00:00 - 22:18:00	GUIKAHSIN 0.00 USD
12/04/2026	Learn Azure Group 3 Section 02 - Online learning Azure	02:16:54 09:00:00 - 11:16:54	Sabrina Heng 0.00 USD

Figure 7: Clockify report